

```

1  #include <stdio.h>
2  int main() {
3      /*
4      1. input n,at,bt
5      2. pid,bt,wt,tat,rt arrays
6      3. rt[i] = bt[i]
7      4. while(completed < n)
8      5. find tat & totals
9      6. print and averages
10     */
11     int n;
12     printf("No of P: \n");
13     scanf("%d", &n);
14
15     int arrival_time[10];
16     int rem_time[10];
17     int burst_time[10];
18     int wait_time[10];
19     int turnaround_time[10];
20
21     for (int i = 0; i < n; i++) {
22         printf("Enter Arrival time of P%d: ", i + 1);
23         scanf("%d", &arrival_time[i]);
24         printf("Enter Burst time of P%d: ", i + 1);
25         scanf("%d", &burst_time[i]);
26         rem_time[i] = burst_time[i];
27     }
28     int completed = 0;
29     int currentTime = 0;
30     while (completed < n) {
31         int minTime = 999;
32         int shortest_job_id = -1;
33         for (int i = 0; i < n; i++) {
34             if (rem_time[i] > 0 && arrival_time[i] <= currentTime &&
35                 rem_time[i] < minTime) {
36                 minTime = rem_time[i];
37                 shortest_job_id = i;
38             }
39         }
40         if (shortest_job_id == -1) {
41             currentTime++;
42             continue;
43         }
44         rem_time[shortest_job_id]--;
45         currentTime++;
46
47         if (rem_time[shortest_job_id] == 0) {
48             completed++;
49             turnaround_time[shortest_job_id] =
50                 currentTime - arrival_time[shortest_job_id];
51             wait_time[shortest_job_id] =
52                 turnaround_time[shortest_job_id] - burst_time[shortest_job_id];
53         }
54     }
55
56     int totalWT = 0;
57     int totalTAT = 0;
58     printf("\n SRTF \n");
59     printf("PID\tAT\tBT\tWT\tTAT\n");
60     for (int i = 0; i < n; i++) {
61         printf("%d\t%d\t%d\t%d\t%d\n", i + 1, arrival_time[i], burst_time[i],
62             wait_time[i], turnaround_time[i]);
63
64         totalWT += wait_time[i];
65         totalTAT += turnaround_time[i];
66     }
67     printf("Average WT: %.2f \n ", (float)totalWT / n);
68     printf("Average TAT: %.2f \n", (float)totalTAT / n);
69 }
70

```